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Review

Sirtuins, epigenetics and longevity

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ABSTRACT

Aging of organisms begins from a single cell at the molecular level. It includes changes related to telomere shortening, cell senescence and epigenetic modifications. These processes accumulate over the lifespan. Research studies show that epigenetic signaling contributes to human disease, tumorigenesis and aging. Epigenetic DNA modifications involve changes in the gene activity but not in the DNA sequence. An epigenome consists of chemical modifications to the DNA and histone proteins without the changes in the DNA sequence. These modifications strongly depend on the environment, could be reversible and are potentially transmittable to daughter cells. Epigenetics includes DNA methylation, noncoding RNA interference, and modifications of histone proteins. Sirtuins, a family of nicotine adenine dinucleotide (NAD+)-dependent enzymes, are involved in the cell metabolism and can regulate many cellular functions including DNA repair, inflammatory response, cell cycle or apoptosis. Literature shows the strong interconnection between sirtuin expression and aging processes. However, the direct relationship is still unknown. Here, we would like to summarize the existing knowledge about epigenetic processes in aging, especially those related to sirtuin expression. Another objective is to explain why some negative correlations between sirtuin activity and the rate of aging can be assumed.

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1. Introduction

Sirtuins were initially described as transcription-silencing histone deacetylases in yeast. Because yeast Sir2 protein is upregulated by calorie restriction, and calorie restriction has a lifespan-extending effect in most species, a positive correlation between Sir2 activity and replicative lifespan in yeast was proved. Furthermore, it has been confirmed that calorie restriction, at least in yeast, extends lifespan indeed through upregulation of sirtuins (Kennedy et al., 1995; Guarente, 2000). Although one later study failed to confirm lifespan regulation in fruit flies by calorie restriction in a sirtuin-dependent manner (Burnett et al., 2011), another, newer study confirmed it (Whitaker et al., 2013). The apparent discrepancy between those two results can be related to some differences in the methods applied, regarding both the calorie restriction regime and general approach towards sirtuin activity regulation (all-tissue regulation vs. tissue-specific and tissue-selective regulation). When taken together, attempts to solve the contradictions mentioned above can introduce a more useful approach towards studying SIRT activity and its effects,

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emphasizing the importance of evaluating SIRT KO or SIRT induction effects in a tissue-dependent manner (Guarente, 2013).

In mammals, there are seven sirtuins (SIRT1 – 7), which have different profiles of actions, substrate affinity, and subcellular compartmentation. Yet, all of them share a similar catalytic domain and use NAD+ as a co-substrate. Although initially identified as deacetylases, sirtuins are now known to have much more kinds of enzymatic activity, including deacylase and O-ADP-ribosylase activity. SIRT3, SIRT4, and SIRT5 are mitochondrial proteins, while SIRT1, SIRT6and SIRT7 are nuclear enzymes, and – as such – can take part in the epigenetic regulation of cell phenotype, especially that they target histones and transcription factors. SIRT2 can be shuttled between nucleus and cytoplasm, depending on the phase of the cell cycle. There is some evidence that SIRT3 can be active not only in the mitochondria, but also in the cell nucleus, especially in absence of stress (Scher et al., 2007), although later studies contradict those results and find SIRT3 as "exclusively mitochondrial" (Cooper and Spelbrink, 2008; ; Hallow et al., 2008). The latter papers state that "finding" SIRT3 in cell nucleus was an artifact which could result from some unrecognized shortcomings in the materials and methods applied. Yet, it is known that SIRT3 expression does affect the activity of some nuclear transcription factors, and thus – gene expression. Hallows et al. attribute it to indirect regulation, through SIRT3-dependent PTMs in intracellular signaling pathways (Hallows et al., 2008). Conflicting results as to the intracellular location of SIRT3 can make the readers aware of how much remains to elucidate – both when trying to list the possible targets of SIRT3, and when trying to distinguish between its direct and indirect regulatory effects upon cellular machinery.

In yeast, in which Sir2 protein was first discovered (Guarente, 2000) the correlation between sirtuin activity and the rate of aging is very significant, especially in reference to replicative aging (Sinclair and Guarente, 1997). In mammals, positive correlations have been observed between the levels of SIRT3 and SIRT6 expression and longevity (Bellizzi et al., 2005; Kanfi et al., 2012). Furthermore, sirtuin knock-out (KO) mice have a reduced lifespan because of accelerated aging (SIRT6 KO, SIRT7 KO) (Kawahara et al., 2009; Vazquez et al., 2016; Vakhrusheva et al., 2008a, 2008b) or because of increased cancer incidence (SIRT3 KO, SIRT4 KO) (Finley and Haigis, 2012; Jeong et al., 2013; Laurent et al., 2013).

Sirtuins prevent DNA damage (DD) and promote DNA damage repair (DDR) acting independently at many different stages. Main DDR-promoting actions of sirtuins include: PARP activation (Mao et al., 2011), glutamine anaplerosis (Jeong et al., 2013) and facilitation of DSB repair through homologous recombination, instead of non-homologous end joining (Kaidi et al., 2010; Huertas 2010). DD-preventing actions of sirtuins may include: suppression of ROS production in the mitochondria (Haigis et al., 2012; Ho et al., 2013) and activation of ROS-neutralizing enzymes (Tao et al., 2010; Qiu et al., 2010; Lin et al., 2013).

Human sirtuin which has been the most extensively studied, and is the most similar to yeast Sir2 in terms of exerted actions and the profile of enzymatic activity, is SIRT1. SIRT1-activating compounds can mimic some beneficial effects of calorie restriction (CR) in mice, especially in mice fed with high-fat diet (Mitchell et al., 2014). SIRT1 can share some metabolic properties of SIRT6–facilitating DDR, suppressing the activity of NF-KB, as well as promoting gluconeogenesis and fatty acid oxidation. In some metabolic contexts in the liver, SIRT1 and SIRT6 can backup one another: low-carbohydrate, high-fat diet can suppress SIRT1, but at the same time, large amounts of FFA-derived acyl groups can activate SIRT6 (Feldman et al., 2013). On the other hand, CR activates both SIRT1 and SIRT6. High-fat, high-carbohydrate diet can suppress both SIRT1 and SIRT6, which is consistent with its observed detrimental effects on hepatocytes (non-alcoholic fatty liver disease) (Nassir and Ibdah, 2016). Central actions of SIRT1 are neuroprotective, and there is some evidence that SIRT1 or SIRT6 induction limited to the brain can extend lifespan in mammals, through suppressing metabolic actions of NF-KB in the hypothalamus (SIRT6) and exerting a general neuroprotective effect (SIRT1) (Kawahara et al., 2009). Upon aging, SIRT1 is recruited to DNA damage sites, which can diminish its activity towards other regions of the genome and result in a well-known "transcriptional noise" as a hallmark of cellular senescence (Oberdoerffer et al., 2008).

Unlike SIRT1, SIRT2 is not neuroprotective, and its increased activity with age is related to higher incidence of neurodegenerative diseases (Outeiro et al., 2007). This striking difference between the effects of SIRT1 and SIRT2 induction in reference to CNS can make the readers aware that mammalian SIRTs should not be simplistically perceived as "orthologues of yeast Sir2, which extend lifespan". Mammalian SIRT2 has shown no lifespan-extending effects so far, and moreover, it tends to be overexpressed with age in neurodegeneration-prone mice (Outeiro et al., 2007).

SIRT3 and SIRT4 improve the efficacy of electron transport chain (ETC) (Haigis et al., 2012; Ho et al., 2013), while SIRT5 can activate a mitochondrial isoform of superoxide dismutase (Cu-SOD) (Lin et al., 2013). Unlike SIRT3 or other sirtuins, SIRT4 is downregulated by caloric restriction (CR) alone, while its main activators include ATM and ATR proteins, in response to DNA damage. An important and unique role of SIRT4 is to block glutamine metabolite entrance to tricarboxylic acid cycle (TCA), which allows the use of glutamine-derived nitrogen for purine nucleotide synthesis during DNA damage repair (Jeong et al., 2013). Because of apparently paradoxical down-regulation of SIRT4 by CR, it is thought to optimize ATP synthesis in response to nutrient deprivation, as well as to provide a retrograde signaling from mitochondria to the cell nucleus. Since AMPK is downregulated by SIRT4 and SIRT4 itself is downregulated by CR, AMPK is actually upregulated by CR to take part in the subsequent upregulation of SIRT1 and SIRT6–this is why it is called a "retrograde" signaling. On the other hand, DNA damage alone, despite its possible inhibitory effect on AMPK in a SIRT4-dependent manner, is unlikely to downregulate SIRT1, SIRT3 and SIRT6 in an AMPK-dependent manner, because in such case, function of sirtuins other than SIRT4 is rescued by a co-existing activation of p53 (Laurent et al., 2013; Jeong et al., 2013; Ho et al., 2013).

SIRT6 is a nuclear sirtuin that shows many activities promoting DDR and genome stability, as well as functioning as TSP. Deacylase activity of SIRT6 refers mainly to H3K18 and H3K56 in specific domains of the genome (Tasselli et al., 2016), while its O-ADP-ribosylation on KAP1 and PARP1 proteins can suppress detrimental actions of L1 retrotransposons and promote DSB repair, respectively (Mao et al., 2011; Van Meter et al., 2014).

SIRT7 has been linked to DNA repair and genome stability regulation (Vazquez et al., 2016). SIRT7 KO mice develop a progeroid syndrome (multi-organ dysfunction, reduced lifespan) while SIRT7 itself seems to promote DSB repair through NHEJ (non-homologous end joining). SIRT7 activity towards nucleolar regions of chromatin accounts for its cell-rejuvenating potential, promoting mitochondrial biogenesis and protein synthesis. Unlike SIRT3 or SIRT6, SIRT7 is often upregulated in cancer cell lines, exactly because of its strongly cytoprotective actions. From the reasons discussed already, it can be interpreted rather as an adaptative evolution of cancer cells than as finding SIRT7 as a possible oncoprotein causative in reference to carcinogenesis, especially when having in mind its DDR-promoting functions.

Recruitment of SIRTs to DNA damage sites upon cellular genotoxic stress can finally contribute to their diminished activity in their original sites of action (Oberdoerffer et al., 2008). This can account for a gradual alteration of cells phenotype in aging organisms, promoting epigenetic changes produced due to diminished SIRT activity as chromatin modifiers. The same can refer to other chromatin modifiers that are recruited to DNA damage sites; the resulting epigenetic rearrangements can be adaptive in a short-term perspective (because of having DNA damage repaired), but detrimental in a long-term perspective (because of a lasting alteration in the pattern of gene expression). On the other hand, unrepaired DNA damage could be detrimental even in a short-term perspective, and this is why SIRT KO mice develop either progeroid phenotypes or increased cancer incidence.

Since most sirtuins (except SIRT4) are activated by moderate undernutrition, the sirtuin-induced metabolic status of the cell and insulin-induced metabolic status of the cell are most often mutually exclusive (Erol, 2010). Moreover, inhibition of crucial downstream effectors of insulin and IGF signaling pathway (IIS), such as mTOR, results both in the inhibition of aging and in bringing SIRT activity to normal level, especially if it has been too low (Kanfi et al., 2012; Erol, 2010). These findings, when taken together, can help to find possible explanations of the beneficial effect of caloric restriction (CR) on lifespan. Caloric restriction-mimicking (CRM) is a branch of experimental medicine where scientists try to achieve some beneficial effects of CR without actually applying CR. So far, both sirtuin activating compounds (STACs) and replenishment of NAD^+ shared the beneficial effects of CR in reference to an endothelial function, glucose homeostasis, condition of the cardiovascular system, the general condition of the connective tissue, and cellular ROS level (Mahajan et al., 2011; Imai and Guarente, 2014).

Having in mind all the reservations and nuances mentioned above, one can still have some doubts as to whether SIRT-dependent loss of cell-senescence phenotype (CSP) would be favorable in mammals. There are some concerns that CSP can protect damaged cells from replication, and thus loss of CSP can make mammals susceptible for cancer (Vaziri et al., 2001). Indeed, in many cancer cells, SIRTs are upregulated, which is then protective for a cancer cell, not for the host. However, overexpression of SIRTs in cancer cells is not universal, and moreover, there is little likelihood that it is a causative factor for carcinogenesis. Firstly, because SIRTs in general strongly promote DNA damage repair, while most types of cancer are caused by accumulated, unrepaired DNA damage. Secondly, even in some types of cancer that seem to result more from epigenetic than genetic alterations, SIRTs are not primary targets of these alterations. Third and foremost, SIRT1, SIRT3, SIRT4 and SIRT6 function much like tumor suppressor proteins (TSPs) in healthy cells. They promote DNA damage repair (DDR) and maintenance of genomic stability independently of promoting DDR. Finally, many sirtuins do activate "real" TSPs (gatekeepers and caretakers) while inactivating oncoproteins – perhaps SIRT3 and SIRT6 are the best examples. In these contexts, mere observation of upregulated SIRTs in some cancer cells does not have to rise concerns about their causative role in carcinogenesis. Many proteins are dysregulated in cancer cells, but it is widely known that carcinogenesis results from abnormal downregulation of TSPs (e.g. p16, Rb, p53, FoxOs, or PTEN) rather than from primary upregulation of cytoprotective enzymes like SIRTs. In other words, although cytoprotection is necessary to survive from the standpoint of already established cancer cell lines, the same cytoprotection alone is far too little to transform a healthy cell into a cancer cell. Thus, when stipulating possible roles of SIRTs in a healthy multicellular organism, one need not have concerns about some issues which are – by definition – absent in a healthy organism, and their incidence is increased by SIRT KO rather than SIRT upregulation (Stunkel and Campbell, 2011).

Although there is some evidence for lifespan-extending effects of using HDAC inhibitors (Samuelson et al., 2007), which is apparently conflicting with finding the same result with upregulation of SIRTs (which function like HDAC themselves), it should be noted that: 1- findings of lifespan extension due to HDAC inhibition pertain to worms, not to mammals, and 2- the inhibited HDAC in worms belong to quite different class of HDAC than sirtuins. Taken together, it can – once again – warn the readers against "projecting" research results interpretation from worms to mammals, and against too simplistic insight into such interpretations. After taking "corrections" for a different class of enzymes affected (despite similar function in terms of biochemical nomenclature) and different organisms studied, the contradictions can disappear.

It is widely known that epigenetic regulation (HDAC) is a downstream effector of sirtuin activity. But can it also work in the opposite direction? Are there some feedback loops involving SIRT activity and epigenetic regulation? The development of organismal aging phenotype in mammals, despite the initial presence of sirtuin "anti-aging" activity, may suggest that either SIRT activity alone is not sufficient to prevent aging or SIRT activity is affected by a complex interplay between stress-related and epigenetic factors, which – taken together – blunts their activity with age. In this paper, we will try to clarify at least some of these hypotheses.

2. Sirtuin activity and histone modifications in epigenetic aging

Histone post-translational modifications (PTMs) are epigenetic mechanisms that control longevity in various organisms via regulation of transcriptional activation or inactivation, chromosome packaging, and DNA damage repair. Best known histone covalent PTMs include: acetylation, methylation, phosphorylation, ADP-ribosylation, ubiquitylation and sumoylation, with acetylation being the most prevalent. Acetylation of histones leads to a more open chromatin structure and is linked to transcriptionally active domains. Histone deacetylation results in gene silencing due to chromatin inaccessibility (Feser and Tyler, 2011). With age, chromatin is thought to undergo a more euchromatic and transcriptionally active conformation in general (Tsurumi and Li, 2012). However, according to some researchers, an increase in de novo heterochromatinisation of certain regions in senescent cells may also occur. An increase in histone acetylation mediated by decreased sirtuin activity contributes to aging phenotype in some organisms (Chandra et al., 2012). Although in various organisms these effects can be different and phenotypically predominant effects may affect different histones (Dang et al., 2009; Kawahara et al., 2009; Sen et al., 2016).

Sirtuin-mediated regulation of gene silencing and DNA repair is achieved through the deacetylation of site-specific lysine residues within histones. SIRT1, SIRT2 and SIRT6 are the most studied proteins in this process (Jing and Lin, 2015; Kanfi et al., 2012). In yeast and *Caenorhabditis elegans* (*C. elegans*) Sir2 deacetylates lysine residues of H4K16 and H3K9. As a result, histones H3 and H4 establish a more repressive conformation of chromatin, which is related to lifespan extension (Jing and Lin, 2015). In aging yeast cells, decreased Sir2 activity is strictly related to the increase in H4 lysine 16 acetylation and loss of histones at specific subtelomeric regions (Imai et al., 2000). Sir2 KO in *Drosophila* results in a greatly increased level of H3K56ac which reconfirms that Sir2 deacetylates H3K56 (Das et al., 2009). Inhibition of SIRT1 in mammalian cells causes hyperacetylation of H3K56 and promotes genomic instability. Furthermore, it was proposed that accumulation of H3K56ac at the S phase, stimulates SIRT1 activity (Yuan et al., 2009). The decrease in Sir2 protein abundance is correlated with enhanced H4K16 acetylation and loss of histones at specific subtelomeric regions in replicatively old yeast cells (Dang et al., 2009). H4K16dac may be crucial to the formation of condensed chromatin, to which SIRT2 is a major contributor during the cell cycle when condensed chromatin must be generated anew (Vaquero et al., 2006). The level of H4 acetylation varies during the cell cycle – it is low in M/G1 phase and increases in S/G2 phase (Vaquero et al., 2006; Contrepois et al., 2012). Some researchers suggest that SIRT2 may be involved in the global decline of H4K16ac levels during the G2/M transition. It was noted that mitotic cells with lower SIRT2 activity had higher levels of H4K16ac (Vaquero et al., 2006; Vaquero 2009). In vitro, SIRT2 preferentially deacetylates histone H4 at Lys16 and SIRT2 KO mouse embryonic fibroblasts display hyperacetylated H4K16 phenotype during mitosis (Haigis and Guarente, 2006), which may suggest that SIRT2 activity is essential for the global deacetylation of H4K16ac during mitosis (Contrepois et al., 2012; Serrano et al., 2013). This is relevant not only because the importance of this mark on mitosis and cell cycle in general, but also because a mark that depends on H4K20me1, H4K20me3 was shown to increase in old mouse (Sarg et al., 2002). Despite being a predominant mitochondrial protein, SIRT3, like SIRT2, takes part in H4K16ac deacetylation when transported to the nucleus under certain conditions (Seto and Yoshida, 2014).

SIRT6 is a nuclear protein and deacetylates H3K9 and H3K56. It has been recently presented to improve longevity in male mice by 16% when overexpressed (Kanfi et al., 2012). SIRT6 is histone H3 lysine 9 deacetylase/deacylase that modulates and protects telomeric chromatin in human cells (Michishita et al., 2008). Moreover, SIRT6 deficiency leads to telomere dysfunction with chromosomal fusions and premature cellular aging. SIRT6 KO mice develop abnormalities that overlap with aging-associated degenerative processes (Mostoslavsky et al., 2006). SIRT6 also interacts with NF-kB and deacetylates H3K9 at NF-kB target gene promoters. Increased H3K9ac in SIRT6-deficient cells is associated with enhanced apoptosis and cellular senescence related to NF-kB-dependent modulation of gene expression pattern (Kawahara et al., 2009). Investigation of histone acetylation status at telomeres in S-phase revealed that SIRT6 is required for the maintenance of low levels of H3K9 acetylation. Moreover, H3K9 was hyperacetylated at telomeric chromatin in SIRT6 KO mouse cells, which provides *in vivo* evidence that SIRT6 plays an important role in deacetylating this histone residue (Michishita et al., 2008; Feldman et al., 2013). SIRT6 has a weak deacetylase activity towards free histones in vitro, while it deacetylates histones H3 and H4 when they are packaged as nucleosomes. SIRT6 activity is nucleosome-dependent, which suggests that its binding to the nucleosome can take part in its activation (Gil et al., 2013). Moreover, recent studies suggested that SIRT6 can also regulate H3K18ac in pericentromeric chromatin which is essential for mitotic fidelity (Barber et al., 2012; Tasselli et al., 2016).

The main substrate of a nuclear protein SIRT7 is the H3 histone, deacetylated by SIRT7 at Lys 18 (H3K18dac) (Barber et al., 2012; Watroba and Szukiewicz, 2016). Deacetylation of the H3 histone at Lys 18 represses gene expression. SIRT7 can be recruited by specific transcription factors (ELK4 and Myc) to deacetylate H3K18 and suppress downstream gene transcription (Tong et al., 2016). The recent identification of a series of hitherto unknown histone acylations, including succinylation, indicates the role of SIRT7 in epigenetic mechanisms. SIRT7 is a histone desuccinylase which is functionally linked to chromatin compaction and genome stability. A recent study has revealed that SIRT7-catalysed H3K122 desuccinylation is crucial for DNA-damage response and cell survival (Li et al., 2016).

Until not long ago, studies on lifespan had mainly focused on histone acetylases and deacetylases, specifically the SIR2 deacetylase family (Dang et al., 2009; Kawahara et al., 2009). It seems that histone methyltransferases and demethylases can also affect longevity (Han and Brunet, 2012). It was observed that the lesser the extent of histone H3K36 methylation, the more abnormal transcription occurred in a subgroup of genes and the shorter was the yeast's lifespan. In contrast, deletion of K36me2/3 demethylase Rph1 increased H3K36me3 within these genes, reduced the abnormal transcription, and extended lifespan by approximately 30% (Sen et al., 2015). Histone 3 lysine 4 trimethylation (H3K4me3) is generally considered to be associated with induced gene expression, and H3K27me3 – with transcriptional repression (Han and Brunet, 2012). It is known that SIRT1 activates histone methyltransferases at their target sites and consecutively promotes histone methylation (Jing and Lin, 2015). For instance, SIRT1 deacetylates SUV39H1 (suppressor of variegation 3–9 homolog 1) and contributes to elevated levels of SUV39H1 activity, which results in increased levels of the H3K9me3 modification (Vaquero et al., 2007), acting as a repressive mark. Moreover, SIRT1 and SUV39H1 are responsible for chromatin modulations associated with facultative heterochromatin (FH) formation. FH is reversible, may possess some markers specific for euchromatin, but still can be used for transcription in a development-dependent manner (Vaquero et al., 2004, 2007). Studies report that deacetylation of H1K26 by SIRT1 allows its coordinated methylation by EZH2 (Enhancer of zeste homolog 2) (Kuzmichev et al., 2004). Upon some senescence-triggering signals, the EZH2 level falls, resulting in the loss of H3K27me3 mark at the INK4A locus, which is thought to be a major factor related to the induction of aging and carcinogenesis (Shumaker et al., 2006). Likewise, decreased levels of H3K27me3 are correlated with a shortened lifespan in *C. elegans* (Ni et al., 2012). In contrast, limiting the H3K4me3 level in *C. elegans* is beneficial for its longevity due to reduced transcription of genes that would normally lead to aging (Han and Brunet, 2012). In *C. elegans*, inactivation of UTX-1 demethylase extended lifespan and increased the global levels of the H3K27me3 mark (Maures et al., 2011).

Another important PTM – phosphorylation is a context-dependent process. Phosphorylated residues may be involved in opposite events such as in chromatin condensation associated with mitosis and meiosis, and in chromatin relaxation linked to transcription activation (H3S10 and H3S28). Therefore, phosphorylation of certain residues should not be considered in isolation, because other PTMs may determine its final effects (Rossetto et al., 2012).

Other PTMs, such as ubiquitylation and sumoylation, include the covalent addition of ubiquitin or SUMO (small ubiquitin-related modifier) to lysine residues within histones and some other proteins (Sen et al., 2016). In human cells, the SAGA (Spt-Ada-Gcn5-Acetyltransferase) deubiquitination activity is essential for full transcriptional activity mediated by nuclear receptors (androgen receptor, AR; glucocorticoid receptor, GR) (Zhao et al., 2008; Wallberg et al., 2000). In yeast, the SAGA histone deubiquitinase module (DUBm) components limit replicative lifespan. Strains lacking genes encoding DUBm, such as Sgf73, are long-lived (McCormick et al., 2014). According to the research, loss of Sgf73 may lead to enhanced Sir2 activity. Sgf73 shortens yeast replicative lifespan through various pathways mediated by Sir2, which lead to maintenance of more compacted chromatin structure (Grillari et al., 2010; Sen et al., 2016). In higher organisms, the role of deubiquitinases is still unclear. Ubiquitylation of DAF-16 which is the ortholog of the FOXO family in *C. elegans* leads to a shorter lifespan in worms (Li et al., 2007). Withal, SUMO expression is increased in old rats and senescent fibroblasts. Overexpression of SUMO is linked to premature aging in worms (Rytinki et al., 2011).

The ADP-ribosylation is a reversible PTM involved in many cellular processes (gene expression, apoptosis, DNA repair, cell signaling). Interestingly, SIRT6 is known of having multiple biochemical activities,dependent not only on protein deacetylation, but also on mono-ADP-ribosylation. Van Meter et al. characterized a new function of SIRT6 that may be relevant in the context of aging. They have found that SIRT6 is a powerful suppressor of L1 retrotransposon activity through mono-ADP-ribosylation of the nuclear co-repressor protein, KAP1 (KRAB-associated protein 1), in mouse embryonic fibroblasts (MEFs) (Van Meter et al., 2014). L1 retrotransposons are quite abundant class of transposable elements, and their activity has been implicated in a variety of age-related disorders (Hancks and Kazazian, 2012). It has also been reported that SIRT6 can mono-ADP-ribosylate Poly (ADP-ribose) polymerase 1 (PARP1) on lysine residue 521, and PARP1 has been implicated in silencing of retrotransposons *in Drosophila* (Mao et al., 2011). Furthermore, SIRT6-dependent activation of PARP1 can enhance DSB repair under oxidative stress.

Overall, the epigenetic mechanisms involving SIRT activity and their effect on histone modifications in aging are still awaiting a full explanation (Table 1). However, the data available from longevity studies conducted so far indicate a strong correlation between histone acetylation and aging.

3. Sirtuins vs. DNA methylation in aging

Despite the evident role of sirtuins in the histone deacetylation/deacylation, they are also involved in other epigenetic mechanisms. DNA methylation is the most studied mechanism responsible for the gene expression regulation and maintenance of genomic stability. In many regions of the genome, including promoters, enhancers or silencers, dinucleotides composed of cytosine and guanine (CpG) occur in high density, mostly in the areas called CpG islands. In CpG pairs, the C is susceptible to being modified by one of DNA methyltransferases (DNMTs) (Mitchell et al., 2016). Generally, DNA methylation in gene promoters is negatively correlated with the gene expression, while methylation in gene bodies attenuates transcription. Some findings demonstrate that promoter-associated sites with overall low methylation tend to gain methylation with age, while those with high DNA methylation lose it with time. However, the pattern of DNA methylation is highly divergent, depending on the tissue type and the presence of epigenetically specific sites that are consistently associated with age (Jones et al., 2015). Studies show that the level of DNA methylation becomes decreased with age, and general DNA hypomethylation may be associated with age-related genome instability and loss of telomere integrity. However, there are some site-specific areas, mainly within the promoter regions, which have been found to become hypermethylated with age. Among them, there are insulin-like growth factor-II (IGFII), estrogen receptor1 (ESR1), hypermethylated in cancer 1 (HIC1), p16/CDKN2A, caspase-8 (CASP8)etc. (Jung and Pfeifer, 2015).

Epigenetic changes in aging can be influenced by different environmental factors, including diet. Literature shows the potential of certain dietary compounds in modulating DNA methylation, which contributes to aging. It is known currently that healthy lifestyle and calorie restriction extend the lifespan. However, the underlying mechanisms are still under investigation. More recently, studies have revealed changes in DNA methylation in overweight human subjects and identified molecules essential in mediating the response to low-calorie supply. Researchers revealed increased expression of SIRT1 in several tissues after calorie restriction. Some genes chosen by in silico analysis, connected with dietary restrictions, showing at the same time altered DNA methylation with age, were affected by SIRT1 concentration in the cells (Ions et al., 2013). Studies based on cell culture experiments show that DNA methylation events are correlated to SIRT1 expression levels and that there is a decrease in SIRT1 expression in aging organisms (Buler, Andersson, and Hakkola 2016). There is a hypothesis that dietary restrictions upregulate SIRT1 transcription, which leads to increased histone deacetylation and increased DNA methylation (Wakeling et al., 2009). SIRT1-mediated effects could increase lifespan through maintenance of epigenomic integrity and proper DNA methylation pattern.

Additionally, the gradual loss of DNA methylation with age may be related to the reduced activity of DNA methyltransferases (DNMTs). SIRT1 upregulates DNMT1 relevant in DNA methylation process. DNMT1 is deacetylated by SIRT1, which improves its DNA methyltransferase activity and its transcription-repressive function (Peng et al., 2011). Hence, the higher SIRT1 expression levels, the greater activity of DNMTs and likelihood of proper DNA methylation patterns, essential in maintaining genomic stability.

Since SIRT1 plays an important role in aging, and at the same time, Alzheimer's disease (AD) is one of the most common age-dependent disorders, the researchers wanted to know if there is a connection between these two processes. The investigation showed significant hypermethylation of SIRT1 gene and demethylation of the β-amyloid precursor protein (APP) gene in AD patients (Hou et al., 2016). SIRT1 expression was inversely correlated with β-amyloid protein production. However, in Parkinson's disease, another age-related disorder, Outeiro et al. revealed that SIRT2 contribute to microtubule destabilization, a process associated with α-synuclein aggregation and creation of Lewy bodies (Outeiro et al., 2007). Thus SIRT2, unlike SIRT1, does not seem to be neuroprotective.

Loss of SIRT1 activity is more likely to be driven in obese patients by different epigenetic mechanisms. Nevertheless, reduced level of SIRT1 can decrease lipolysis, promote fat accumulation, and increase obesity-related inflammation. However, neither SIRT1 nor SIRT7 genes were regulated by DNA methylation in obese patients regardless of age, despite lower levels of SIRT1 in these patients when compared to normal-weight individuals (Kurylowicz et al., 2016). As inflammatory processes increase with age and inflammatory diseases coincide with an age-related increase in DNA methylation (Jung and Pfeifer, 2015), it may be expected that obese patients may eventually engage additional epigenetic mechanisms, such as DNA methylation, and begin the aging process earlier. DNA hypermethylation in the inflamed tissues is also associated with genes recognized by the Polycomb Complex, a part of global silencing system regulating stem cell development (Wakeling et al., 2015). The polycomb group proteins (PcG) form complexes, bind to their target genes (PCGTs) and act as repressors. The hypermethylation of PCGTs is present both in senescent and cancer cells. The age-related DNA methylation involves produced *de novo* DNA methyltransferases (DNMTs) and regulates the accessibility for Polycomb repressive complexes (PRCs). Hence, genes that were originally silenced by PRCs are now methylated, which constitutes an alternative silencing mechanism. However, this epigenetic shift reduces the plasticity of some regulatory programs and can impair the growth potential of the cells. There is a hypothesis that SIRT1 affects DNA methylation at PCGTs, although the mechanism remains unknown (Wakeling et al., 2009).

Besides SIRT1, SIRT6 and SIRT7 are also present in the cell nucleus, and they may also have a direct effect on transcription and expression levels of genes related to aging. Scientists discovered decreased levels of both SIRT6 and SIRT7 in old rat liver tissue in comparison to younger ones, which was consistent with decreased levels of these sirtuins in senescent endothelial cells (Lee et al., 2014). However, it is still unclear if these sirtuins affect DNA methylation processes associated with aging.

4. Sirtuins and micro RNA in aging

Micro RNA (miRNA) comprises an array of short (19–25 nucleotide length) particles that may pair with complementary messenger RNA molecules, thus regulating gene expression at the level of protein synthesis. Expression profile of miRNA is changing over the lifespan. It is an open question whether these changes are the result of aging or a factor driving towards senility. Over recent years, several reports have shown that this epigenetic mechanism might influence aging (Cosentino and Mostoslavsky, 2013). Factors that modulate sirtuin expression can include miRNAs. Only

Table 1
The effects of sirtuins on histone activity and consequences of their insufficiency.

Sirtuin	Histone modifications	Effects	Deficiency
SIRT1	H4-K16dac H3-K9dac H1-K26dac H3-K14dac H-K16dac H3-K56dac	Transcriptional silencing, genome stability Transcriptional silencing Cell-cycle regulation, DNA repair	Genomic instability, DNA damage accumulation in case of co-existing genotoxic stress
SIRT1	H3-K79me3[a] H3-K27me3[a]	Genome stability[a]	Accelerated cellular aging[a]
SIRT2	H3-K56dac H4-K16dac	Chromatin condensation mitosis, DNA repair	Cell cycle inhibition at G_2/M checkpoint, loss of histones at specific subtelomeric regions
SIRT2	H4K20me1[a]	Genome stability, S-phase progression[a]	Genomic instability and chromosomal aberrations, tumorigenesis[a]
SIRT3	H4-K16dac[a]	Chromatin silencing, transcriptional silencing of stress genes, DNA repair[a]	DNA damage[a]
SIRT6	H3-K9dac H3-K56dac	Protection of subtelomeric chromatin, Promoting DNA repair and genome stability, Prevention of aging and neurodegenerative diseases via silencing the NF- kB promoter sites	Aging-associated degenerative diseases, Inflammaging, excessive cell apoptosis, lethal hypoglycemia
SIRT7	H3-K18dac	Facilitation of DSB repair through NHEJ, Promotion of genomic stability Induction of rRNA transcription	Proliferation stop, accelerated organismal aging, DNA damage, impaired glucose and lipid metabolism

[a] Indirect modifications.

SIRT1 is regulated by numerous miRNAsincluding: miR-9,miR-34a, miR-132, miR-181a, miR-195, miR-199a, miR-199b and miR-204 (Tabuchi et al., 2012). Unfortunately, many of these interactions have not yet been investigated in details.

The best-studied relationship between miRNAs and sirtuins involved in aging is the positive feedback regulatory loop involving miR-34a, SIRT1 and p53 (Yamakuchi and Lowenstein, 2009). The p53 tumor suppressor protein is called a guardian of the genome because, in response to DNA damaging factors, it triggers cellular responses like cell cycle arrest, apoptosis, or cellular senescence. On one hand, increased p53 activity inhibits SIRT1 expression, while on the other hand –it can enhance miR-34a production. In turn,miR-34a molecules bind to 3‘UTR of SIRT1 mRNA and suppress its translation. The decrease in SIRT1 synthesis increase p53 acetylation and its activity, which can increase miR-34a production and finally lead to apoptosis (Yamakuchi and Lowenstein, 2009). The miR-34a expression is supposed to affect the lifespan of *C. elegans* and *Drosophila* (de Lencastre et al., 2010; Liu et al., 2012). Experiments performed on human cell cultures show that SIRT1 can inhibit cell aging through deacetylation of p53, and thus -inhibition of its proapoptotic activity (Vaziri et al., 2001). Studies carried out on mice suggest that miR-34a levels in mouse PBMCs and plasma may be suitable biomarkers of age-related changes in the brain. The increase in miR-34a level inversely correlates with expression of SIRT1 both in blood and brain samples (Li et al., 2011b). Increased expression of miR-9 was also observed in rat liver and HEK293 cell cultures. Both some studies on animal organs and those on HEK293 cell line show that miR-9 expression correlates with a decreased level of SIRT1. Similar observations have been reported for another miRNA (miR-93) (Li et al., 2011a). miR-34a also suppresses translation of other genes like nicotinamide phosphoribosyltransferase (NAMPT), which in turn limits intracellular NAD+ level, and thus modifies SIRT1 enzymatic activity and transcription (Buler, Andersson, and Hakkola 2016). Because miR-34a can regulate other sirtuins, like SIRT6 and -7 (Zhang et al., 2015; Lefort et al., 2013), it is possible that similar regulatory loops exist for other sirtuins (Buler et al., 2016).

Another aging-related process that involves sirtuins and miRNAs is regulation of stem cell metabolism. With advancing age,stem cells lose their ability to produce progenitors, as well as their differentiation capacity. Influence of sirtuins on pluripotency and differentiation is not entirely understood, but it was shown that SIRT1, SIRT2, SIRT3, SIRT6 and SIRT7 modulate these processes (Correia et al., 2016). The best-studied sirtuin is SIRT1 which is required for proper differentiation of both embryonic and adult stem cells. SIRT1 influence on cell differentiation was studied in cultures of mesenchymal, neural and hematopoietic stem cells, as well as inSIRT1 KO animals (Correia et al., 2016). It was observed that SIRT1 expression in stem cells is regulated by numerous miRNAs: miR-34a miR-181a and b, miR-9, miR-204, miR-199b, and miR-135a (Saunders et al., 2010; Yu et al., 2015). SIRT1 was also identified as a target for miR-199a which is a regulator of phenotypic switch involved in vascular cells differentiation in endothelial cells (Li et al., 2015).

Aging is related to decreased level of sex hormones, impaired memory, and increased anxiety. One of the genes related to memory and aging is SIRT1 gene. Age-related changes may be driven by miRNAs. In menopause rat model, SIRT1 expression is inhibited by miR-9 (Rao et al., 2013). It was shown that SIRT1 mRNA translation in the hippocampus of young animals is lower than in older ones and inversely correlates with miR-9 expression. Another two miRNAs that substantially change with age are miR-7a and miR-495. It was observed that level of miR-7a is linked to the level of circulating 17β-estradiol (E2). Treatment of rats with E2 completely abolished the age-related effect on miR-7a (Rao et al., 2013).

In elderly people, increased risk of neurodegenerative disorders is observed. It was shown that dysregulation of miR-181 is critically involved in the development of AD. This miRNA regulates expression of c-Fos and SIRT1. An experiment performed in a mouse model of AD showed that miR-181 upregulation in the hippocampus can decrease c-Fos and SIRT1 expression, and thus reduce synaptic plasticity (Rodriguez-Ortiz et al., 2014).

5. Summary

We can conclude that sirtuins expression is negatively correlated with aging and that numerous epigenetic modifications may be driven by sirtuins (Fig. 1). The most obvious role of sirtuins in the epigenetic regulation of cell function is their histone deacetylating and deacylating function (HDAC). Sirtuin-mediated

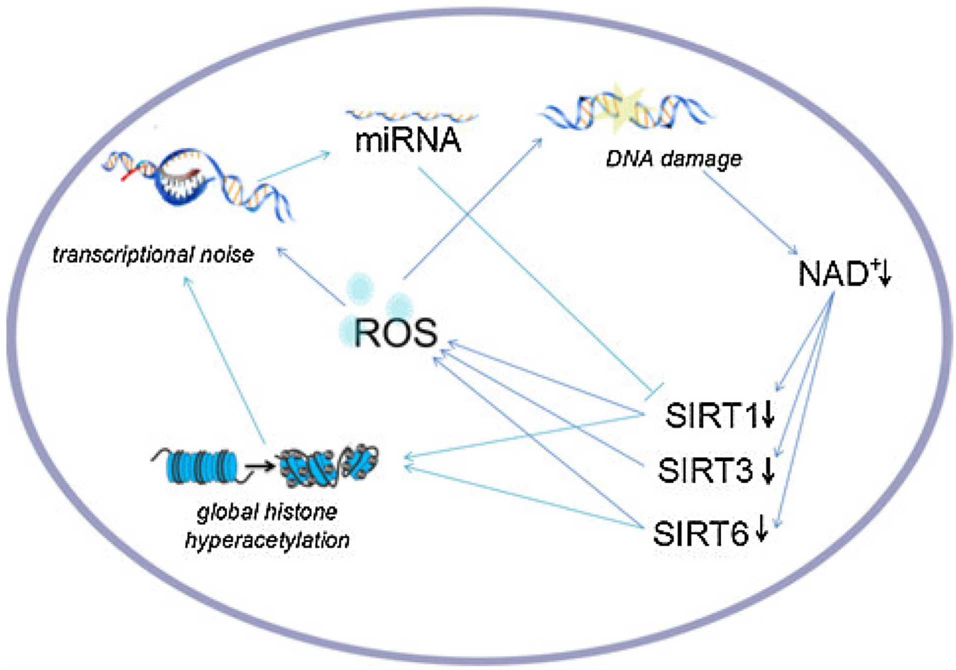

Fig. 1. Possible mechanisms that may contribute to the decrease in sirtuin expression inside the cells. The loss of cellular NAD+ in response to the DNA damage leads to the decreased expression of sirtuins. The depletion in sirtuin level leads to the generation of ROS and histone hyperacetylation in cells. These, in turn, are involved in transcriptional noise generation and the creation of miRNAs. The described events can create a positive feedback loop responsible for the sirtuin downregulation in the course of aging.

regulation of gene silencing and DNA repair is achieved through the deacetylation of site-specific lysine residues within histones. Deacetylation of histones H4K16, H3K9, and H3K56 by sirtuins contributes to establishing a more repressive conformation of chromatin, which is related to lifespan extension. It is suggested that increased histone acetylation, which occurs in old cell populations, is associated with decreased sirtuins activity. Histone methylation can also affect the lifespan. Specific histone marks such as e.g. H3K4me3 and H3K27me3 represent different chromatin states, opened and condensed respectively. H3K4me3 is linked to aging phenotype while H3K27me3 – to lifespan extension. It is known that sirtuins activate histone methyltransferases and consecutively promote histone methylation. The role of sirtuins in ubiquitylation and sumoylation is less understood.

The studies revealed that the level of DNA methylation decreases with age, which may result in age-related genome instability and loss of telomere integrity. Since the research confirmed that there is also a decrease in SIRT1 expression in aging organisms, we can assume that DNA methylation events may be correlated with SIRT1 expression levels. Age-related DNA hypomethylation is accompanied by a decrease in DNA methyltrasferases activity. DNMT1 is upregulated by SIRT1 through deacetylation. Hence, the higher SIRT1 concentration, the greater activity of DNA methyltransferases necessary for DNA methylation. We can conclude that sirtuins, especially SIRT1, enhance DNA methylation processes essential in the maintenance of genomic stability and longevity.

Relatively least-known epigenetic modifications are related to miRNAs. Recent studies show that dysregulated miRNAs and SIRTs can mediate the development of neurodegenerative diseases, affect the metabolism of stem cells and the process of menopause. Positive feedback regulatory loop involving miR-34a, SIRT1 and p53 has been widely studied, and it is suggested that similar loops can exist for other sirtuins. The interplay between sirtuins and various miRNAs can decrease stem cells ability to produce progenitors and decrease stem cell capacity to differentiate. Menopause-associated correlations between SIRT1, miR-9 and levels of sex hormones have also been observed.

Aging involves a series of events including changes related to telomere shortening, cell senescence and modifications in the epigenome. Since sirtuins' activity is age-dependent we may assume the interplay between sirtuin expression and the rate of epigenetic changes in senescent cells. However, the question concerning negative correlations between sirtuin activity and the rate of aging remains open. Likewise, the possibility to apply sirtuin-activating compounds in the longevity-promoting therapy requires further investigations.

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